HUJI Applicative Research Report — Healthcare — Week 34, 2026

3 high-potential paper(s) · Healthcare · Weekly · Week 34, 2026 · Generated August 17, 2026

Hebrew University researchers are advancing groundbreaking solutions across critical sectors. New insights into cellular uptake mechanisms promise to revolutionize gene therapy delivery, making treatments more accessible and efficient. In AgriTech, innovative genetic discoveries are paving the way for climate-resilient wheat, addressing global food security challenges. Furthermore, smart gene therapy approaches are being developed to precisely target neurological conditions, minimizing side effects and enhancing therapeutic outcomes.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Zvi Peleg** — AgriTech, Genomics, Synthetic Biology
- **Tawfeeq Shekh-Ahmad** — Drug Discovery, Neuroscience, Genomics, Clinical

HEALTHCARE — RESEARCH HIGHLIGHTS

1. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake. 33/50

"Novel MCP PNA enables simple, cost-effective intracellular drug delivery, transforming gene and antisense therapies."

Main researcher: Eylon Yavin

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Published in ACS polymers Au · 2026-06

ABOUT THIS RESEARCH

Researchers developed a new type of synthetic DNA analogue called MCP PNA that can naturally enter cells without needing complex delivery vehicles. This solves a major historical bottleneck where these stable molecules were too large or poorly charged to cross cell membranes.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty 6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential 8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size 9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness 3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength 7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

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2. Antagonistic pleiotropy at the stem solidness 1 locus balances wheat stem architecture and yield components under terminal drought. 31/50

"Genetic breakthrough creates climate-resilient, high-yield wheat, securing global food supply against drought."

Main researcher: Zvi Peleg · zvi.peleg@mail.huji.ac.il.

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Published in TAG. Theoretical and applied genetics. Theoretische und angewandte Genetik · 2026 Aug 16

ABOUT THIS RESEARCH

Researchers identified a specific genetic locus (stem solidness 1) in wheat that controls the trade-off between stem structural strength and grain yield. The study explains how adjusting this genetic trait can help wheat plants survive terminal drought without sacrificing overall productivity.

COMMERCIAL ANGLE

Development of climate-resilient, high-yield wheat cultivars through precision breeding or CRISPR gene editing targeting the s1 locus.

COMMERCIALISATION METRICS

Novelty 5/10

The identification of a specific locus controlling stem architecture and its trade-off with yield under drought is a valuable incremental contribution, but follows established paradigms of QTL mapping and antagonistic pleiotropy in crop science.

Commercial Potential 7/10

The identification of a specific locus controlling the trade-off between stem strength and yield under drought provides a direct target for marker-assisted breeding or CRISPR editing in wheat, a global staple crop. While highly relevant for seed commercialization, the score is tempered by the inherent complexity of 'antagonistic pleiotropy' which may require precise tuning rather than a simple binary switch.

Market Size 9/10

Wheat is one of the world's most critical staple crops; a genetic solution to maintain yield under terminal drought addresses a massive global food security market with high industrial demand.

Tech Readiness 3/10

The research focuses on identifying a genetic locus and the biological trade-off (antagonistic pleiotropy) between stem architecture and yield, which is fundamental discovery research (TRL 2-3) rather than a developed seed product or breeding protocol ready for commercial deployment.

IP Strength 7/10

The identification of a specific genetic locus (stem solidness 1) governing a trade-off between architecture and yield provides a clear target for Marker-Assisted Selection (MAS) or CRISPR editing, which is highly patentable in agricultural biotech.

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3. Activity-dependent antioxidant gene therapy protects against epileptogenesis. 31/50

"Smart gene therapy prevents epilepsy by activating only during brain activity, minimizing side effects."

Main researcher: Tawfeeq Shekh-Ahmad · Tawfeeq.Shekh-Ahmad@mail.huji.ac.il.

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Published in Signal transduction and targeted therapy · 2026-06

ABOUT THIS RESEARCH

This research describes a gene therapy approach that uses natural brain activity to trigger the production of antioxidant enzymes. This mechanism aims to prevent the development of epilepsy following an initial brain injury.

COMMERCIAL ANGLE

Development of smart, responsive gene therapies for neurological disorders that activate only during specific physiological states to minimize side effects.

COMMERCIALISATION METRICS

Novelty 7/10

The work is highly innovative for shifting the therapeutic paradigm from reactive seizure control to proactive, activity-dependent gene modulation to prevent epileptogenesis. However, while the mechanism is sophisticated, antioxidant gene therapy is an established concept, making the novelty incremental rather than a complete disruption of biological principles.

Commercial Potential 7/10

The technology targets a high-unmet-need clinical indication (epileptogenesis) with a specific mechanism (antioxidant gene therapy), offering a clear path toward a specialized therapeutic product. However, the lack of abstract details regarding delivery vectors and scalability limits a higher score.

Market Size 7/10

The target market for epilepsy therapeutics is massive and high-value, representing a multi-billion dollar global neurology segment. However, the score is tempered by the specific niche of epileptogenesis prevention, which is a subset of the broader epilepsy market and faces complex clinical adoption hurdles.

Tech Readiness 3/10

The title suggests a fundamental mechanistic study using gene therapy in a disease model, which typically implies preclinical, lab-based proof-of-concept research rather than a developed therapeutic candidate.

IP Strength 7/10

The use of activity-dependent promoters for targeted gene therapy provides a strong basis for composition-of-matter and method-of-use patents, though defensibility depends on the novelty of the specific promoter/transgene combination.

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